

VARUN SHARMA

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EXPERIENCE

Software Developer — Tata Consultancy Services (TCS)

Dec 2023 – Present

- Engineered and deployed scalable Node.js/Express microservices and **REST APIs** for a high-traffic healthcare application, enhancing system modularity and backend reliability.
- Automated infrastructure provisioning on AWS (EC2, ECS, Lambda, S3) using **CloudFormation**, reducing manual deployment errors and ensuring environment consistency.
- Optimized **ElasticSearch** clusters for a 500k+ record healthcare dataset, reducing search latency by 40% through custom analyzers and **NLP** driven Query DSL.
- Established real-time monitoring and alerting via AWS **CloudWatch**, maintaining 99.9% uptime for mission critical patient data services.
- Collaborated in an Agile environment with cross functional teams to translate business requirements into technical specifications, ensuring on-time delivery of core features.

Full Stack Developer Intern — MRIIC

Jan 2023 – June 2023

- Developed full-stack features using the **MERN** stack and Django, delivering end-to-end functionality for internal web applications.
- Designed and integrated **RESTful APIs**, optimizing data retrieval and ensuring secure communication between frontend interfaces and cloud hosted databases.
- Built responsive user interfaces using **Bootstrap** and JavaScript, resulting in a seamless cross-device experience and improved user retention metrics.
- Participated in Agile sprints, collaborating with senior developers to troubleshoot technical bottlenecks and deploy scalable features within tight project deadlines.

PROJECTS

Vector-Based Product Retrieval System

[Link](#) | [GitHub](#)

NLP, Vector Embeddings, OpenSearch, Node.js, Transformers

- Engineered a Semantic Search Engine leveraging all-mpnet-base-v2 embeddings and OpenSearch K-NN to enable "intent-based" product discovery beyond literal keyword matching.
- Developed a Full-Stack Web Application (Node.js/Express) that translates natural language queries into 768-dimension vectors for real-time similarity mapping.
- Optimized Search Relevance by implementing a custom OpenSearch index schema, effectively mapping abstract concepts (e.g., "corporate commute") to relevant product categories like formal wear.

Chanakya School Landing Page

[Link](#) | [GitHub](#)

Built a school management web application using Django.

- Architected and deployed a full stack management portal on **PythonAnywhere**, enabling centralized administration of student records and academic data.
- Automated communication workflows by integrating **Django Mailer**, reducing manual notification efforts and improving update delivery speed.
- Optimized frontend accessibility using **Bootstrap 5**, achieving 100% mobile responsiveness and cross-browser compatibility.

TECHNICAL SKILLS

- Languages & Scripting:** Java, JavaScript
- Frameworks & Libraries:** Node.js, Express, React.js
- Cloud & Databases:** AWS, Opensearch
- Concepts:** Data Structures & Algorithms (DSA), Problem Solving

EDUCATION

Manav Rachna International University

Faridabad, Haryana

B.tech CSE

2019 – 2023

CERTIFICATIONS AND PARTICIPATIONS

- Basics of Kubernetes (KODEKLOUD): [Link](#).
- Basics of Docker (KODEKLOUD): [Link](#).
- Learn Figma - UI/UX Design Essential Training : [Link](#).
- Mentored junior developers on Django fundamentals and full-stack web application development during the Samsung Django Hackathon (May 2023).
- Performed guitar in music events.
- Represented MRIIRS in table tennis.